

Jiahui Yang . Jarvis Consulting

Early-career AI and Machine Learning Engineer with a Master's degree in Electrical and Computer Engineering and a Bachelor's degree in Computer Science. Skilled in Python, SQL, machine learning, and data engineering, with hands-on experience building end-to-end data pipelines and predictive modeling solutions. Experienced in scalable data processing and cloud-based data platforms to deliver reliable, efficient, and data-driven solutions. Proven ability to collaborate with cross-functional teams, with a strong focus on data quality, scalability, and continuous learning.

Skills

Proficient: Python, Linux/Bash, Machine Learning, ETL, RDBMS/SQL, Agile/Scrum, Git

Competent: Apache Spark, Databricks, Microsoft Azure, Scikit-learn, Docker

Familiar: Generative AI/LLMs, AWS, Apache Airflow, Apache Kafka, Apache Hadoop

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_JiahuiYang

Cluster Monitor [GitHub]:

- Designed and implemented a Linux-based cluster monitoring agent to collect and analyze system metrics across multiple hosts.
- Developed Bash scripts to extract hardware specifications (CPU, memory, disk) and real-time usage data, and stored structured data in a PostgreSQL database for centralized monitoring.
- Containerized the database using Docker for portability and simplified deployment.
- Automated data collection using cron jobs, enabling continuous tracking of system performance and resource utilization.

Python Data Analytics [GitHub]:

- Performed data analytics on London Gift Shop (LGS) retail transaction data using Python, Pandas, and NumPy to uncover customer purchasing patterns and support targeted marketing strategies.
- Cleaned and transformed retail sales data by handling missing values, filtering cancelled transactions, and calculating revenue metrics for analysis.
- Applied RFM customer segmentation and created Matplotlib visualizations to identify high-value customers and generate insights that could benefit the business.

Spark [GitHub]:

- Performed data analytics using PySpark on Apache Zeppelin running on a GCP Dataproc Hadoop cluster, leveraging Spark SQL and distributed DataFrame processing.
- Built an ETL pipeline following the Medallion Architecture (Bronze, Silver, and Gold) in Azure Databricks to ingest, clean, and analyze fraud detection data, create dashboards from gold tables, and orchestrate automated workflows using Databricks Jobs.
- Built a DLT pipeline following the Medallion Architecture (Bronze, Silver, and Gold) in Databricks to automate stock market data ingestion, transformation, dashboard reporting, and workflow orchestration.

Home Credit Default Risk: End-to-End Credit Scoring System [GitHub]:

- Built an end-to-end credit risk prediction system using the Home Credit dataset with over 300,000 loan applications.
- Performed EDA, feature engineering, customer segmentation, model training, hyperparameter tuning, SHAP explainability, and PSI-based model monitoring.
- Compared Logistic Regression, Random Forest, and Gradient Boosting models using AUROC, Gini, KS, and AUPRC metrics.
- Developed a production-ready Scikit-learn Pipeline and generated model validation documentation following financial industry best practices.

Highlighted Projects

End to End Azure Data Engineering Project [GitHub]: Designed and implemented an end-to-end data pipeline on Azure to ingest, transform, and analyze data for business reporting. Built data ingestion workflows using Azure Data

Factory to extract data from SQL Server and store it in ADLS Gen2 following a medallion architecture. Transformed and optimized data using Azure Databricks, PySpark, and Delta Lake, and enabled analytics through Azure Synapse serverless SQL. Developed interactive Power BI dashboards and integrated Azure Key Vault and Microsoft Entra ID for secure data access and pipeline management.

E-Hospital Medical Chatbot System: Developed and implemented an ETL pipeline for an E-Hospital chatbot system to process and structure medical data from multiple sources. Extracted and transformed XML-based medical data into structured CSV files using Python and Pandas, ensuring data quality through validation and cleaning. Collaborated with a cross-functional team to integrate the chatbot into an existing platform, enabling efficient access to healthcare information. Delivered project presentations and a promotional demo to communicate the system's functionality and impact.

Professional Experiences

Jr. AI/ML Engineer, Jarvis (Mar 2026-present): Worked in an Agile environment following GitFlow-based development workflows to build end-to-end data engineering, analytics, and machine learning solutions using Python, SQL, Linux, Apache Spark, Databricks, Docker, and cloud platforms. Developed ETL pipelines, scalable data processing workflows, and machine learning applications that support data-driven and AI-enabled solutions.

Data Analytics Internship, Invision Trading (May 2025-Dec 2025): Performed end-to-end data processing and feature engineering on financial datasets of over 5,000 mutual fund schemes using Python (Pandas), including data cleaning, normalization, and preparation for analysis. Developed a data-driven scoring model to identify top-performing funds based on return and risk metrics. Designed interactive Power BI dashboards to visualize key financial indicators and support data-driven investment decision-making.

Education

University of Ottawa (2022-2025), Master of Engineering, Electrical and Computer Engineering

Nanjing University of Information Science and Technology (2018-2022), Bachelor of Engineering, Computer Science and Technology

Miscellaneous

- Databricks Certified Data Engineer Associate
- Microsoft Certified: Azure Data Engineer Associate
- IBM Data Engineering Professional Certificate
- Cooking and exploring new recipes
- Driving and road trips
- Regular gym training and fitness